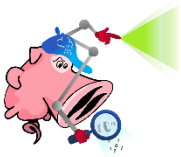


Snort Challenge - Live Attacks

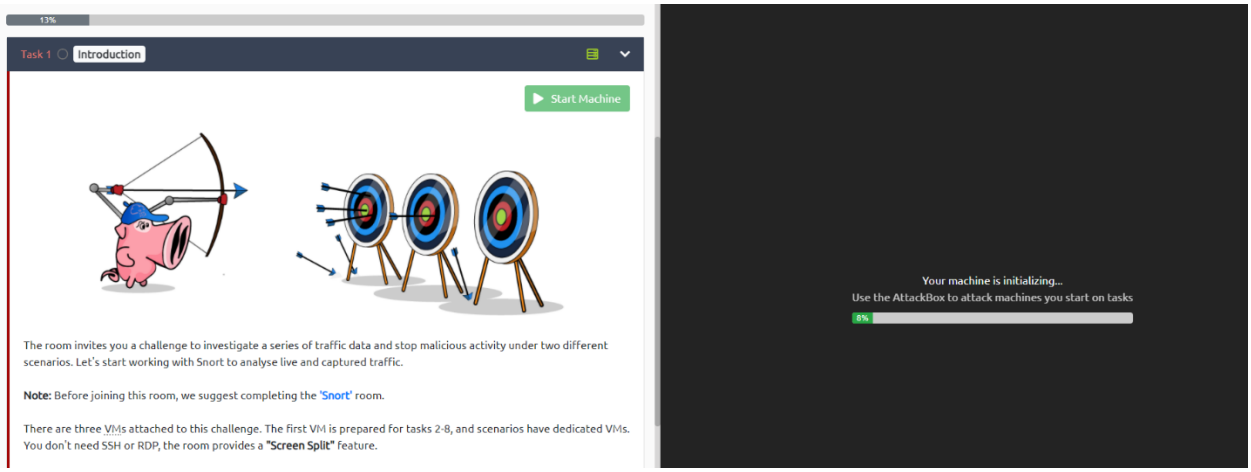
Put your snort skills into practice and defend against a live attack

Task 1 – Introduction

The room invites you to a challenge where you will investigate a series of traffic data and stop malicious activity under two different scenarios. Let's start working with Snort to analyse live and captured traffic.

Before joining this room, we suggest completing the Snort room.

Note: There are two VMs attached to this challenge. Each task has dedicated VMs. You don't need or , the room provides a "Screen Split" feature.



Answer the questions below

Read the task above.

No answer needed.

Task 2 – Scenario 1 | Brute-Force

Use the attached to finish this task.

[+] THE NARRATOR

J&Y Enterprise is one of the top coffee retails in the world. They are known as tech-coffee shops and serve millions of coffee lover tech geeks and IT specialists every day.

They are famous for specific coffee recipes for the IT community and unique names for these products. Their top five recipe names are;

WannaWhite, ZeroSleep, MacDown, BerryKeep and CryptoY.

J&Y's latest recipe, "Shot4J", attracted great attention at the global coffee festival. J&Y officials promised that the product will hit the stores in the coming months.

The super-secret of this recipe is hidden in a digital safe. Attackers are after this recipe, and J&Y enterprises are having difficulties protecting their digital assets.

Last week, they received multiple attacks and decided to work with you to help them improve their security level and protect their recipe secrets.

This is your assistant J.A.V.A. (Just Another Virtual Assistant). She is an -driven virtual assistant and will help you notice possible anomalies. Hey, wait, something is happening...

[+] J.A.V.A.

Welcome, sir. I am sorry for the interruption. It is an emergency. Somebody is knocking on the door!

[+] YOU

Knocking on the door? What do you mean by "knocking on the door"?

[+] J.A.V.A.

We have a brute-force attack, sir.

[+] THE NARRATOR

This is not a comic book! Would you mind going and checking what's going on! Please...

[+] J.A.V.A.

Sir, you need to observe the traffic with Snort and identify the anomaly first. Then you can create a rule to stop the brute-force attack. GOOD LUCK!

```
ubuntu[redacted]:~$ snort -V

   ,,_     -*> Snort! <*-
  o"  )~   Version 2.9.7.0 GRE (Build 149)
   ''''    By Martin Roesch & The Snort Team: http://www.snort.org/contact#team
           Copyright (C) 2014 Cisco and/or its affiliates. All rights reserved.
           Copyright (C) 1998-2013 Sourcefire, Inc., et al.
           Using libpcap version 1.9.1 (with TPACKET_V3)
           Using PCRE version: 8.39 2016-06-14
           Using ZLIB version: 1.2.11
```

Answer the questions below

First of all, start Snort in sniffer mode and try to figure out the attack source, service and port.

Then, write an IPS rule and run Snort in IPS mode to stop the brute-force attack. Once you stop the attack properly, you will have the flag on the desktop!

Here are a few points to remember:

- **Create the rule and test it with "-A console" mode.**
- **Use "-A full" mode and the default log path to stop the attack.**
- **Write the correct rule and run the Snort in IPS "-A full" mode.**
- **Block the traffic at least for a minute and then the flag file will appear on your desktop.**

Stop the attack and get the flag (which will appear on your Desktop)

```
ubuntu[redacted]:~$ sudo snort -dev -l .
Running in packet logging mode

        --== Initializing Snort ==--
Initializing Output Plugins!
Log directory = .
pcap DAQ configured to passive.
Acquiring network traffic from "eth0".
Decoding Ethernet

        --== Initialization Complete ==--

   ,,_     -*> Snort! <*-
  o"  )~   Version 2.9.7.0 GRE (Build 149)
   ''''    By Martin Roesch & The Snort Team: http://www.snort.org/contact#team
           Copyright (C) 2014 Cisco and/or its affiliates. All rights reserved.
           Copyright (C) 1998-2013 Sourcefire, Inc., et al.
           Using libpcap version 1.9.1 (with TPACKET_V3)
           Using PCRE version: 8.39 2016-06-14
           Using ZLIB version: 1.2.11
```

Here I started Snort in packet logging mode to watch live traffic on the host and save the captured packets locally. This step is used to observe the attack activity, identify the source IP, and determine which service and port are being targeted before writing a rule to block it.

```
ubuntu@                    ~$ ls
Desktop  Documents  Downloads  Music  Pictures  Public  Templates  Videos  snort.log.1774462147
```

Snort successfully captured the observed traffic and saved it to a log file for later analysis.

```
ubuntu@                    :~$ sudo snort -r snort.log.1774462147 | tail
Running in packet dump mode

        --== Initializing Snort ==--
Initializing Output Plugins!
pcap DAQ configured to read-file.
Acquiring network traffic from "snort.log.1774462147".

        --== Initialization Complete ==--

   ,,_     -*> Snort! <*-
  o"  )~   Version 2.9.7.0 GRE (Build 149)
   ''''    By Martin Roesch & The Snort Team: http://www.snort.org/contact#team
           Copyright (C) 2014 Cisco and/or its affiliates. All rights reserved.
           Copyright (C) 1998-2013 Sourcefire, Inc., et al.
           Using libpcap version 1.9.1 (with TPACKET_V3)
           Using PCRE version: 8.39 2016-06-14
           Using ZLIB version: 1.2.11
```

```
===========================================================================
Snort exiting
***A**** Seq: 0xC8C249D8  Ack: 0x522BE2F9  Win: 0x1EB  TcpLen: 32
TCP Options (3) => NOP NOP TS: 1884573991 4119681421
=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+

03/25-18:09:30.609749 10.10.245.36:46610 -> 10.10.140.29:22
TCP TTL:64 TOS:0x0 ID:64291 IpLen:20 DgmLen:64 DF
***AP*** Seq: 0xC8C249D8  Ack: 0x522BE2F9  Win: 0x1EB  TcpLen: 32
TCP Options (3) => NOP NOP TS: 1884573991 4119681421
=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+
```

The traffic shows 10.10.245.36 connecting to 10.10.140.29:22, which suggests repeated access attempts against the SSH service on port 22 and makes 10.10.245.36 the likely brute-force source.

```
ubuntu@ip-10-112-163-102:~$ sudo snort -r snort.log.1774462147 'src 10.10.245.36 and dst port 22'
Running in packet dump mode

        --== Initializing Snort ==--
Initializing Output Plugins!
Snort BPF option: src 10.10.245.36 and dst port 22
pcap DAQ configured to read-file.
Acquiring network traffic from "snort.log.1774462147".

        --== Initialization Complete ==--

   ,,_      -*> Snort! <*-
  o"  )~    Version 2.9.7.0 GRE (Build 149)
   ''''     By Martin Roesch & The Snort Team: http://www.snort.org/contact#team
            Copyright (C) 2014 Cisco and/or its affiliates. All rights reserved.
            Copyright (C) 1998-2013 Sourcefire, Inc., et al.
            Using libpcap version 1.9.1 (with TPACKET_V3)
            Using PCRE version: 8.39 2016-06-14
            Using ZLIB version: 1.2.11

Commencing packet processing (pid=2125)
WARNING: No preprocessors configured for policy 0.
03/25-18:09:07.365596 10.10.245.36:46460 -> 10.10.140.29:22
TCP TTL:64 TOS:0x0 ID:54900 IpLen:20 DgmLen:52 DF
***A**** Seq: 0x6F35BBE7  Ack: 0x6B8C5885  Win: 0x1E1  TcpLen: 32
TCP Options (3) => NOP NOP TS: 1884548141 4119655574
=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+

WARNING: No preprocessors configured for policy 0.
03/25-18:09:07.393281 10.10.245.36:46474 -> 10.10.140.29:22
TCP TTL:64 TOS:0x0 ID:24076 IpLen:20 DgmLen:52 DF
***A**** Seq: 0xBBE7BD9  Ack: 0x1FC50573  Win: 0x1EB  TcpLen: 32
TCP Options (3) => NOP NOP TS: 1884548149 4119655582
=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+

WARNING: No preprocessors configured for policy 0.
03/25-18:09:07.602567 10.10.245.36:46474 -> 10.10.140.29:22
TCP TTL:64 TOS:0x0 ID:24077 IpLen:20 DgmLen:948 DF
***AP*** Seq: 0xBBE7BD9  Ack: 0x1FC50573  Win: 0x1EB  TcpLen: 32
TCP Options (3) => NOP NOP TS: 1884548149 4119655582
=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+
```

The filtered packet log confirmed repeated traffic from 10.10.245.36 to 10.10.140.29 on port 22, indicating a likely SSH brute-force attempt.

```
sudo gedit /etc/snort/rules/local.rules
Open  +   local.rules  /etc/snort/rules   Save
1 # $Id: local.rules,v 1.11 2004/07/23 20:15:44 bmc Exp $
2 # ----------------
3 # LOCAL RULES
4 # ----------------
5 # This file intentionally does not come with signatures.  Put your local
6 # additions here.
7
8 drop tcp 10.10.245.36 any -> 10.10.140.29 22 (msg:"Possible SSH brute-force"; sid:1000001; rev:1;)
```

The rule was updated to drop SSH traffic from the suspected attacker IP so Snort can block the brute-force attempt.

```
ubuntu@[redacted]:~$ sudo snort -Q --daq afpacket -A full -q -u snort -g snort -c /etc/snort/snort.conf -i eth0:eth1
ERROR: log_tcpdump: Failed to open log file "/var/log/snort/snort.log.1774463914": Permission denied
Fatal Error, Quitting..
ubuntu@[redacted]:~$ sudo chown -R snort:snort /var/log/snort
ubuntu@[redacted]:~$ sudo snort -Q --daq afpacket -A full -q -u snort -g snort -c /etc/snort/snort.conf -i eth0:eth1
```

```
ubuntu@ip-[redacted]:~$ cat Desktop/flag.txt
THM{81b7fef657f8aaa6e4e200d616738254}
```

After correcting the inline interface setup and log permissions, Snort was run in IPS mode with full logging, which successfully blocked the SSH brute-force traffic and generated the flag on the Desktop.

Answer: THM{81b7fef657f8aaa6e4e200d616738254}

What is the name of the service under attack?

Answer: SSH

What is the used protocol/port in the attack?

Answer: TCP/22

Task 3 - Scenario 2 – Reverse-Shell

Use the attached to finish this task.

[+] THE NARRATOR

Good Job! Glad to have you in the team!

[+] J.A.V.A.

Congratulations sir. It is inspiring watching you work.

[+] You

Thanks team. J.A.V.A. can you do a quick scan for me? We haven't investigated the outbound traffic yet.

[+] J.A.V.A.

Yes, sir. Outbound traffic investigation has begun.

[+] THE NARRATOR

The outbound traffic? Why?

[+] YOU

We have stopped some inbound access attempts, so we didn't let the bad guys get in. How about the bad guys who are already inside? Also, no need to mention the insider risks, huh? The dwell time is still around 1-3 months, and I am quite new here, so it is worth checking the outgoing traffic as well.

[+] J.A.V.A.

Sir, persistent outbound traffic is detected. Possibly a reverse shell...

[+] YOU

You got it!

[+] J.A.V.A.

Sir, you need to observe the traffic with Snort and identify the anomaly first. Then you can create a rule to stop the reverse shell. GOOD LUCK!

Answer the questions below

First of all, start Snort in sniffer mode and try to figure out the attack source, service and port.

Then, write an IPS rule and run Snort in IPS mode to stop the brute-force attack. Once you stop the attack properly, you will have the flag on the desktop!

Here are a few points to remember:

- **Create the rule and test it with "-A console" mode.**
- **Use "-A full" mode and the default log path to stop the attack.**
- **Write the correct rule and run the Snort in IPS "-A full" mode.**
- **Block the traffic at least for a minute and then the flag file will appear on your desktop.**

Stop the attack and get the flag (which will appear on your Desktop)

```
ubuntu@ip                :~$ sudo snort -dev -l .
Running in packet logging mode

        --== Initializing Snort ==--
Initializing Output Plugins!
Log directory = .
pcap DAQ configured to passive.
Acquiring network traffic from "eth0".
Decoding Ethernet

        --== Initialization Complete ==--

   ,,_      -*> Snort! <*-
  o"  )~    Version 2.9.7.0 GRE (Build 149)
   ''''     By Martin Roesch & The Snort Team: http://www.snort.org/contact#team
            Copyright (C) 2014 Cisco and/or its affiliates. All rights reserved.
            Copyright (C) 1998-2013 Sourcefire, Inc., et al.
            Using libpcap version 1.9.1 (with TPACKET_V3)
            Using PCRE version: 8.39 2016-06-14
            Using ZLIB version: 1.2.11
```

```
ubuntu@                    :~$ ls -lt snort.log*
-rw------- 1 root root 674452 Mar 25 18:58 snort.log.1774465069
ubuntu@ip-                 :~$ sudo snort -r snort.log.1774465069 -n 20
Exiting after 20 packets
Running in packet dump mode

        --== Initializing Snort ==--
Initializing Output Plugins!
pcap DAQ configured to read-file.
Acquiring network traffic from "snort.log.1774465069".

        --== Initialization Complete ==--

   ,,_     -*> Snort! <*-
  o"  )~   Version 2.9.7.0 GRE (Build 149)
   ''''    By Martin Roesch & The Snort Team: http://www.snort.org/contact#team
           Copyright (C) 2014 Cisco and/or its affiliates. All rights reserved.
           Copyright (C) 1998-2013 Sourcefire, Inc., et al.
           Using libpcap version 1.9.1 (with TPACKET_V3)
           Using PCRE version: 8.39 2016-06-14
           Using ZLIB version: 1.2.11

Commencing packet processing (pid=1770)
03/25-18:57:49.177405 10.10.196.55:54128 -> 10.10.144.156:4444
TCP TTL:64 TOS:0x0 ID:11385 IpLen:20 DgmLen:60 DF
******S* Seq: 0x56C8766D  Ack: 0x0  Win: 0xF507  TcpLen: 40
TCP Options (5) => MSS: 8961 SackOK TS: 2358293307 0 NOP WS: 7
=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+

WARNING: No preprocessors configured for policy 0.
03/25-18:57:49.182522 10.112.100.96:58692 -> 10.112.170.211:80
TCP TTL:127 TOS:0x0 ID:21742 IpLen:20 DgmLen:100 DF
***AP*** Seq: 0xB6E0FED0  Ack: 0x21F900BF  Win: 0x1BB  TcpLen: 32
TCP Options (3) => NOP NOP TS: 2959980356 3552844220
=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+

03/25-18:57:49.182607 10.112.170.211:80 -> 10.112.100.96:58692
TCP TTL:64 TOS:0x0 ID:34407 IpLen:20 DgmLen:52 DF
***A**** Seq: 0x21F900BF  Ack: 0xB6E0FF00  Win: 0x1DC  TcpLen: 32
TCP Options (3) => NOP NOP TS: 3552844274 2959980340
=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+
```

```
WARNING: No preprocessors configured for policy 0.
03/25-18:57:49.188369 10.112.100.96:58692 -> 10.112.170.211:80
TCP TTL:127 TOS:0x0 ID:21743 IpLen:20 DgmLen:84 DF
***AP*** Seq: 0xB6E0FF00  Ack: 0x21F900BF  Win: 0x1BB  TcpLen: 32
TCP Options (3) => NOP NOP TS: 2959980361 3552844274
=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+

WARNING: No preprocessors configured for policy 0.
03/25-18:57:49.189240 10.10.144.156:4444 -> 10.10.196.55:54128
TCP TTL:64 TOS:0x0 ID:0 IpLen:20 DgmLen:60 DF
***A**S* Seq: 0x2FE7575C  Ack: 0x56C8766E  Win: 0xF4B3  TcpLen: 40
TCP Options (5) => MSS: 8961 SackOK TS: 1980461006 2358293307 NOP WS: 7
=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+

03/25-18:57:49.199801 10.10.196.55:54128 -> 10.10.144.156:4444
TCP TTL:64 TOS:0x0 ID:11386 IpLen:20 DgmLen:52 DF
***A**** Seq: 0x56C8766E  Ack: 0x2FE7575D  Win: 0x1EB  TcpLen: 32
TCP Options (3) => NOP NOP TS: 2358293307 1980461006
=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+

WARNING: No preprocessors configured for policy 0.
03/25-18:57:49.222785 10.112.100.96:58692 -> 10.112.170.211:80
TCP TTL:127 TOS:0x0 ID:21744 IpLen:20 DgmLen:100 DF
***AP*** Seq: 0xB6E0FF20  Ack: 0x21F900BF  Win: 0x1BB  TcpLen: 32
TCP Options (3) => NOP NOP TS: 2959980396 3552844274
=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+

03/25-18:57:49.222810 10.112.170.211:80 -> 10.112.100.96:58692
TCP TTL:64 TOS:0x0 ID:34408 IpLen:20 DgmLen:52 DF
***A**** Seq: 0x21F900BF  Ack: 0xB6E0FF50  Win: 0x1DC  TcpLen: 32
TCP Options (3) => NOP NOP TS: 3552844314 2959980361
=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+

03/25-18:57:49.223480 10.10.196.55:54128 -> 10.10.144.156:4444
TCP TTL:64 TOS:0x0 ID:11387 IpLen:20 DgmLen:134 DF
***AP*** Seq: 0x56C8766E  Ack: 0x2FE7575D  Win: 0x1EB  TcpLen: 32
TCP Options (3) => NOP NOP TS: 2358293332 1980461006
=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+
```

The captured traffic showed a persistent outbound TCP connection from 10.10.196.55 to 10.10.144.156 on port 4444, which strongly suggested a reverse shell.

```
ubuntu@ip-10-112-170-211:~$ sudo snort -r snort.log.1774465069 'host 10.10.196.55 and host 10.10.144.156 and port 4444'
Running in packet dump mode

        --== Initializing Snort ==--
Initializing Output Plugins!
Snort BPF option: host 10.10.196.55 and host 10.10.144.156 and port 4444
pcap DAQ configured to read-file.
Acquiring network traffic from "snort.log.1774465069".

        --== Initialization Complete ==--

   ,,_     -*> Snort! <*-
  o"  )~   Version 2.9.7.0 GRE (Build 149)
   ''''    By Martin Roesch & The Snort Team: http://www.snort.org/contact#team
           Copyright (C) 2014 Cisco and/or its affiliates. All rights reserved.
           Copyright (C) 1998-2013 Sourcefire, Inc., et al.
           Using libpcap version 1.9.1 (with TPACKET_V3)
           Using PCRE version: 8.39 2016-06-14
           Using ZLIB version: 1.2.11

Commencing packet processing (pid=1978)
03/25-18:57:49.177405 10.10.196.55:54128 -> 10.10.144.156:4444
TCP TTL:64 TOS:0x0 ID:11385 IpLen:20 DgmLen:60 DF
******S* Seq: 0x56C8766D  Ack: 0x0  Win: 0xF507  TcpLen: 40
TCP Options (5) => MSS: 8961 SackOK TS: 2358293307 0 NOP WS: 7
=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+

WARNING: No preprocessors configured for policy 0.
03/25-18:57:49.189240 10.10.144.156:4444 -> 10.10.196.55:54128
TCP TTL:64 TOS:0x0 ID:0 IpLen:20 DgmLen:60 DF
***A**S* Seq: 0x2FE7575C  Ack: 0x56C8766E  Win: 0xF4B3  TcpLen: 40
TCP Options (5) => MSS: 8961 SackOK TS: 1980461006 2358293307 NOP WS: 7
=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+

03/25-18:57:49.199801 10.10.196.55:54128 -> 10.10.144.156:4444
TCP TTL:64 TOS:0x0 ID:11386 IpLen:20 DgmLen:52 DF
***A**** Seq: 0x56C8766E  Ack: 0x2FE7575D  Win: 0x1EB  TcpLen: 32
TCP Options (3) => NOP NOP TS: 2358293307 1980461006
=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+

03/25-18:57:49.223480 10.10.196.55:54128 -> 10.10.144.156:4444
TCP TTL:64 TOS:0x0 ID:11387 IpLen:20 DgmLen:134 DF
```

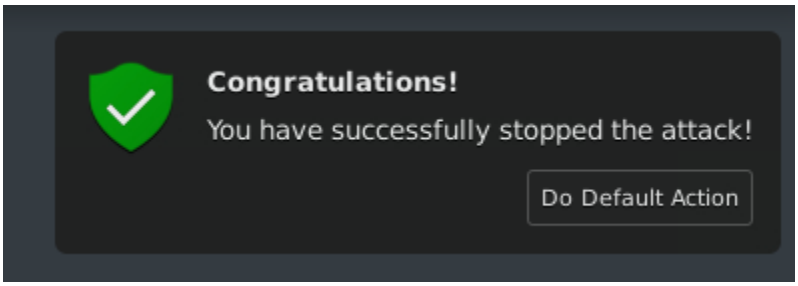
Filtering the capture confirmed repeated bidirectional TCP traffic between 10.10.196.55 and 10.10.144.156 over port 4444, strengthening the assessment that this was an active reverse-shell session.

```
sudo gedit /etc/snort/rules/local.rules
Open  +   local.rules  /etc/snort/rules   Save
1 # $Id: local.rules,v 1.11 2004/07/23 20:15:44 bmc Exp $
2 # ----------------
3 # LOCAL RULES
4 # ----------------
5 # This file intentionally does not come with signatures.  Put your local
6 # additions here.
7
8 drop tcp 10.10.196.55 any -> 10.10.144.156 4444 (msg:"Drop reverse shell"; sid:1000002; rev:1;)
```

```drop tcp 10.10.196.55 any -> 10.10.144.156 4444 (msg:"Drop reverse shell"; sid:1000002; rev:1;)```

A Snort drop rule was created to block outbound TCP traffic from the compromised host to the suspected reverse-shell listener on port 4444.

```
ubuntu@                    :~$ sudo snort -Q --daq afpacket -A full -q -u snort -g snort -c /etc/snort/snort.conf -i eth0:eth1
```



Snort was run in inline IPS mode with full logging, and the custom drop rule successfully blocked the reverse-shell traffic, causing the flag to appear on the Desktop.

```
ubuntu@ip-10-112-170-211:~$ cat Desktop/flag.txt
THM{0ead8c494861079b1b74ec2380d2cd24}
```

Answer: flag: THM{0ead8c494861079b1b74ec2380d2cd24}

What is the used protocol/port in the attack?

Answer: TCP/4444

Which tool is highly associated with this specific port number?

Answer: Metasploit